

Uniform and Automated Deployment in SCVU

Bootc for Air-Gapped, Reproducible Builds

By SCT Benson Teo

Problem

How do you ensure uniform software deployment across machines in an air-gapped environment without spending hours configuring each one?

Solution

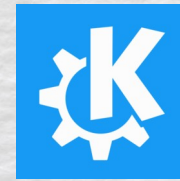
Bootc (Bootable Containers)

Bootc combines OS + Applications + Configs into **1** image that can be **atomically deployed** to any workstation.

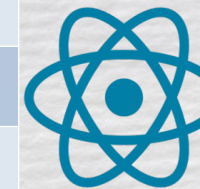
Key Features

- Immutable Root Filesystem
- Container-based builds (OCI)
- Rollback and Updates
- Fast and Reproducible Deployments

Workflow



Stack Breakdown	
Stack(s)	Application(s)
Core OS & Desktop	Fedora 43 & KDE
Development	Python, Git, Docker
Graphics & Design	CUDA, Blender
Windows Emulation	Wine
Office	LibreOffice
File Sharing	Samba, Avahi
Cloud & Containers	Openshift, Kubernetes
JS/Frontend	React, Vue, Typescript



Value Proposition

- Reproducible (same system, every time)
- Offline-capable (ideal for SCVU)
- Fast Deployment
- Customizable (add/change in Containerfile)
- Reduced support (uniform and standard)

Limitations

- Large ISO file size Size: 15.2 GiB (16,277,268,480)
- Certain hardware (read: old) may have issues.
- Immutable Root = Rebuild for changes
- Offline packages become stale
- Linux/Container learning curve

Next Steps

- Pilot with incoming trainees
- Image Variants
- Repository upgrades
- Prep for unit wide deployment

Conclusion

Using Bootc allows us to deploy systems quicker, more consistently and more efficiently.